

Network Devices

Network devices are components used to connect computers or other electronic devices together so that they can share files or resources.

End Devices

A source or destination device in a networked system. These devices form the interface between the human network and the underlying communication network.

- Computers, laptops, file servers, web servers
- Network printers
- VoIP phones
- Security cameras
- Mobile handheld devices

Switches



Switches work the same way as hubs, but they can identify the intended destination of the information that they receive, so they send that information to only the computers that are supposed to receive it. Switches can send and receive information at the same time, so they can send information faster than hubs can. If your home network has four or more computers, or you want to use your network for activities that require passing a lot of information between computers (such as playing network games or sharing music), you should probably use a switch instead of a hub. Switches cost a little more than hubs.

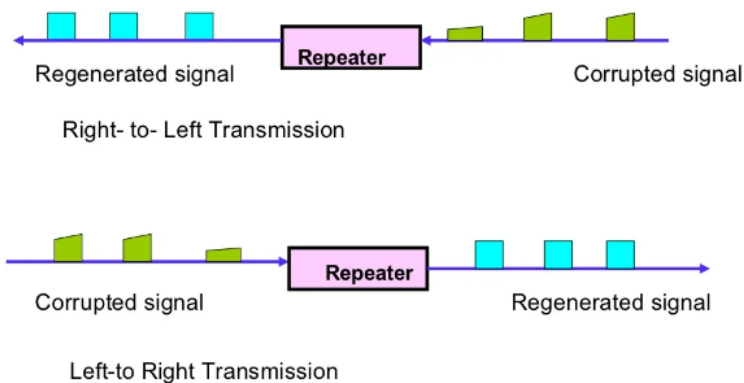
Hubs



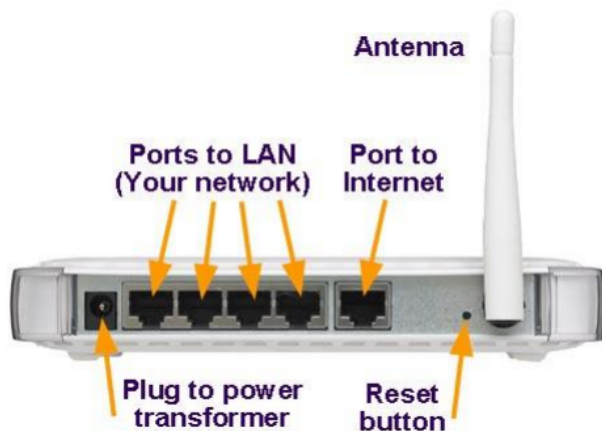
Hubs enable computers on a network to communicate. Each computer plugs into the hub with an Ethernet cable, and information sent from one computer to another passes through the hub. A hub can't identify the source or intended destination of the information it receives, so it sends the information to all of the computers connected to it, including the one that sent it. A hub can send or receive information, but it can't do both at the same time. This makes hubs slower than switches. Hubs are the least complex and the least expensive of these devices.

Repeaters

A common problem in the networking world is that of weakening electrical signals. Electrical signals traveling through wires (such as copper wires used in most networks), weaken due to the wire's electrical resistance. This effect limits the lengths of the cable that can be used. A repeater will overcome this limit, when there is a need to connect two computers at a larger distance. A repeater is connected to two cable segments. Any electrical signal reaching the repeater from one segment, will be regenerated and retransmitted to the other segment.



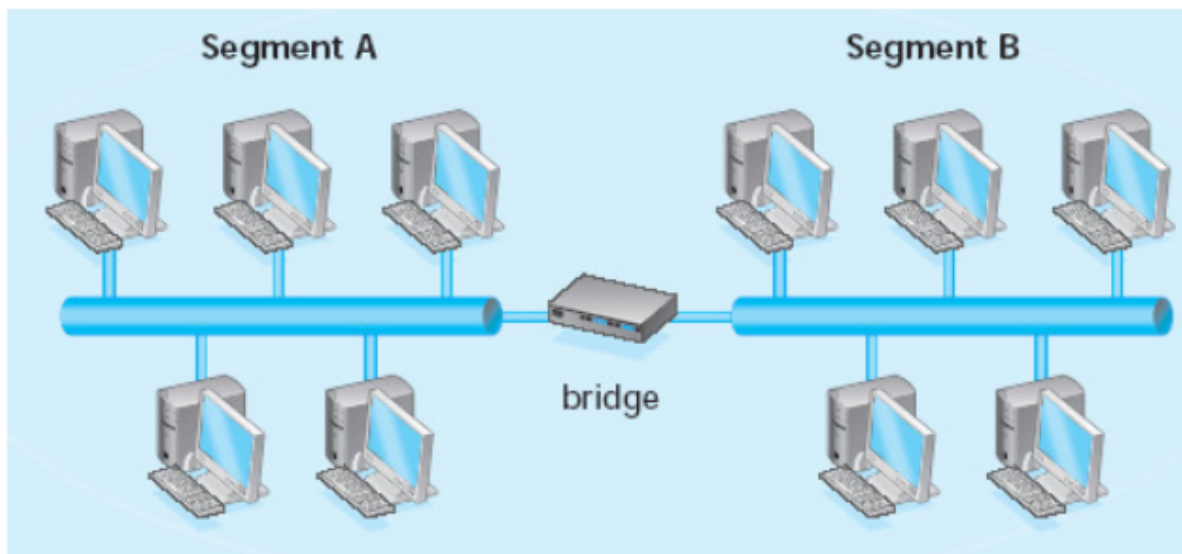
Routers



Routers enable computers to communicate and they can pass information between two networks—such as between your home network and the Internet. This capability to direct network traffic is what gives the router its name. Routers can be wired (using Ethernet cables) or wireless. If you just want to connect your computers, hubs and switches work well; however, if you want to give all of your computers access to the Internet using one modem, use a router or a modem with a built-in router. Routers also typically provide built-in security, such as a firewall. Routers are more expensive than hubs and switches.

Bridges

A device that connects two local-area networks (LANs), or two segments of the same LAN that use the same protocol, such as Ethernet or Token-Ring. A bridge connects networks and forwards frames from one network to another.

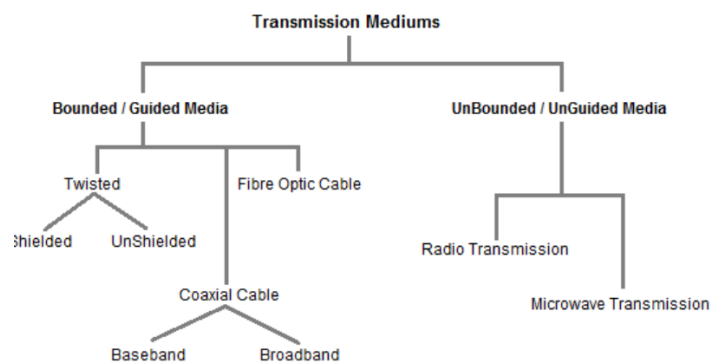


- Information only crosses the bridge if they are addressed for a host on the other side (selective forwarding).
- No hosts on Segment B will receive information from host to host communication on Segment A as they will be blocked by the bridge.
- Bridges are usually used on RG58 networks and are not commonly used today except in wireless networks.

Transmission Media

Transmission media is a pathway that carries the information from sender to receiver.

We use different types of cables or waves to transmit data. Data is transmitted normally through electrical or electromagnetic signals.



Shielded Twisted Pair Cable

The STP or Shielded Twisted Pair is a twisted pair cable that is additionally insulated by foil or/and braided copper shielding. This additional shielding is able to prevent external electromagnetic signals from affecting the signal traveling over the cable.

- Increase performance
- Ensure maximum bandwidth
- Higher cost
- Cable is heavier

Unshielded Twisted Pair Cable

UTP or Unshielded Twisted Pair is not using any additional shielding, except for the plastic sheath around the cable.

- Decreases the cost of the cable
- Less reliable in noise polluted areas
- Reduce performance

Straight through and Crossover cables

Straight-through cable is a type of twisted pair copper wire cable for local area network (LAN) use for which the RJ-45 connectors.

It is identical to crossover cable, except that in the latter the wires on the cable are crossed over so that the receive signal pins on the connector on one end are connected to the transmit signal pins on the connector on the other end.

Ethernet cables can be wired as straight through or crossover. A straight through cable is a type of twisted pair cable that is used in local area networks to connect a computer to a network hub such as a router. Crossover Ethernet cable is more commonly used to connect a computer to a computer and may be a little harder to find since they aren't used nearly as much as straight through Ethernet cable.

Use straight through Ethernet cable for the following cabling:

- Switch to router
- Switch to PC or server
- Hub to PC or server

Use crossover cables for the following cabling:

- Switch to switch
- Switch to hub
- Hub to hub
- Router to router
- Router Ethernet port to PC NIC

- PC to PC

